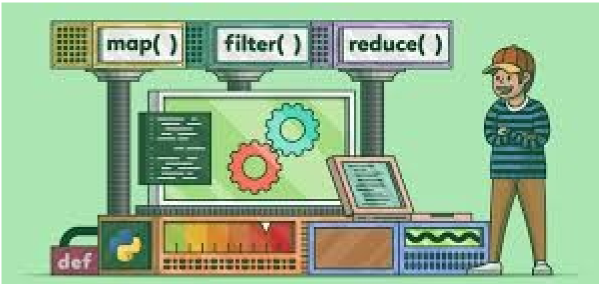
Functional Programming

5CM524



Lab Instructions for 2 hours of labs

Assessed Lab 4

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Aims

The attached instructions form content for 2 hours of labs. **This lab is assessed.**

In this lab you will address three distinct tasks, including a larger task focused on using `map()`, `filter()`, and `reduce()` in an IoT context (like monitoring temperatures in a zoo). They are all looking at using functional concepts in Python.

Assessment and Submission

You will be assessed in the lab, so you need to show your work to staff in the lab. They will complete a marksheet as before and you will then need to upload the marksheet plus your python source code file(s) to course resources under the submission point for the lab assessment will be completed and assessed in the lab, but you have to upload your work and the marking sheet photo to the submission point before the deadline. We will then add marks to Blackboard profile at the start of the following week.

Total Mark: 10

Basic Functional Programming Techniques

In this task, you will practice the basic functional programming concepts, such as lambda functions, `map()`, `filter()`, and `reduce()`.

Task 1: Write a lambda function `is_odd` that checks if a given number is odd. **(1 mark)**

Task 2: Using map() and filter() write a function to convert a list of temperatures from Celsius to Fahrenheit and only return value below 32F. The conversion is as follows: F = (C * 9/5) + 32. **(1 mark)**

IoT Temperature Data Processing for a Zoo

This task simulates processing data from temperature sensors in different areas of a zoo. The data includes temperatures taken from various enclosures, and you will perform a variety of functional programming operations to analyse this data.

You are given a list of temperature readings from various animal enclosures in a zoo. The list includes both normal temperature readings and anomalous readings (such as sensor errors or extreme outliers). You need to process this data using functional programming concepts.

Data Set: The following data represents temperatures (in Celsius) from various animal enclosures in a zoo over a 24-hour period:

```
zoo_temperatures = {
        'lion': [22, 23, 21, 20, 19, 23, 25, 24],
        'elephant': [30, 32, 35, 34, 33, 31, 30, 29],
        'penguin': [5, 6, 7, 6, 5, 4, 4, 5],
        'giraffe': [27, 28, 29, 30, 31, 30, 32, 33],
        'zebra': [21, 22, 23, 22, 21, 20, 19, 20]
}
```

Task 3: You suspect that temperatures below 10°C or above 35°C are erroneous readings. Write a function `filter_anomalies()` that removes all such readings across the entire dataset using `filter()`. **(1 mark)**

Task 4: Write a function `average_temperature()` that computes the average temperature for each animal enclosure, using `map()` and `reduce()`. **(1 mark)**

Real-time Temperature Monitoring System for Zoo Enclosures

You are working on a real-time temperature monitoring system. Each sensor provides a list of temperatures at different time intervals. Your goal is to process this data to identify trends, anomalies, and statistics that help keep the zoo animals safe. You will develop a real-time monitoring system for temperature sensors in a zoo using `map()`, `filter()`, and `reduce()`. This task simulates the continuous processing of incoming data from temperature sensors installed in different animal enclosures.

Data Set: The incoming temperature data is a list of readings for the past 12 hours from multiple sensors. Each sensor provides data as a tuple with the enclosure name and the list of hourly temperature readings.

```
sensor_data = [
    ('lion', [22, 23, 24, 21, 20, 21, 22, 23, 24, 25, 22, 21]),
    ('elephant', [29, 30, 32, 31, 32, 33, 33, 34, 35, 34, 33, 32]),
    ('penguin', [5, 6, 5, 6, 6, 6, 5, 5, 4, 4, 4, 5]),
    ('giraffe', [30, 31, 32, 32, 33, 34, 34, 33, 32, 31, 30, 31]),
    ('zebra', [20, 21, 22, 23, 21, 20, 19, 20, 21, 22, 21, 20])
]
```

Task 5: Write a function `calculate_hourly_averages()` that calculates the average temperature for each enclosure over the 12-hour period. Use `map()` and `reduce()` where appropriate. **(2 mark)**

Task 6: A temperature spike occurs when the difference between two consecutive readings is greater than 5°C. Write a function `identify_temperature_spikes()` that returns a list of enclosures with temperature spikes. **(2 marks)**

Task 7: Write a function `find_hottest_enclosure()` that uses `map()` and `reduce()` to find the enclosure with the highest average temperature. **(2 marks)**